DIVA: Certified Dynamic Individual–Global Max Value Aggregation for Decentralized Multi-Agent Reinforcement Learning

author names withheld

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Abstract

Value factorization has emerged as one of the leading approaches in Multi-Agent Reinforcement Learning (MARL), yet existing theoretical foundations remain restricted to structurally static mixing architectures. Notably, while the Individual–Global Max (IGM) principle is foundational to decentralized execution in MARL, its formal guarantees have thus far been limited to fixed value factorization schemes. In this work, we propose Dynamic Individual–Global Max Value Aggregation (DIVA), a rigorous theoretical framework for certifying stability of the novel abstract class of state-dependent mixing functions. DIVA introduces a state-adaptive gating mechanism that dynamically parameterizes the value aggregation function based on the global state, adjusting agent influence in response to environmental context. It further provides formal verification for this class of dynamic value mixers, deriving a minimal set of sufficient structural conditions (Lipschitz continuity, strict positivity, and monotonicity) that certify decentralized optimality, including guarantees for Dynamic IGM Consistency and Lipschitz Robustness. By formalizing certifiable dynamic value factorization, this work unites architectural flexibility with rigorous control guarantees, extending MARL theory to dynamically reconfigurable mixing functions.

Keywords: Dynamic Multi-Agent Reinforcement Learning, Value Factorization, Individual-Global Max condition

1. Introduction

Cooperative multi-agent reinforcement learning (MARL) remains a widely used framework for complex coordination problems, spanning from autonomous driving to multi-robot system control (Palanisamy, 2020; Orr and Dutta, 2023). Notably, most modern approaches operate under the Centralized Training with Decentralized Execution (CTDE) paradigm (Amato, 2024; Oliehoek et al., 2016), which allows the agents to leverage global information during the training phase while ensuring that during execution each agent can make decisions based solely on local observations.

Under CTDE, coordination is commonly achieved through value factorization, where the joint action–value function Q_{tot} is decomposed into per-agent utilities q_i. This approach originated with Value Decomposition Networks (VDN) (Sunehag et al., 2018), which assume additive separability of agent utilities, and was significantly advanced by QMIX (Rashid et al., 2018), which employs a monotonic mixing network to enable more expressive yet tractable factorizations.

QMIX further popularized a requirement underlying value factorization methods known as the Individual–Global Max (IGM) principle, which ensures consistency between centralized training and decentralized execution. IGM requires that the joint action maximizing the global action–value function Q_{tot} can be recovered by independently maximizing each agent's local utility q_i. This property guarantees that decentralized greedy action selection yields the globally optimal joint action learned during centralized training. While IGM enables scalable coordination without explicit

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communication, it also imposes structural constraints on the form of the value decomposition, dictating both the expressivity and limitations of existing factorization-based MARL methods.

The literature has established various mixing functions satisfying these conditions, but their aggregation operators remain structurally fixed with respect to the global state s. While prior work may employ state-conditioned parameters within a fixed aggregation operator, we introduce aggregation operators that are themselves explicitly conditioned on the state. Specifically, we study a novel class of dynamic mixers in which the form of the value aggregation function is reconfigured as a function of the environmental state. Our framework, Dynamic Individual–Global Max Value Aggregation (DIVA), ensures that this operator-level flexibility remains consistent with the IGM principle through a set of certifiable structural constraints. Our contributions are summarized as follows:

- **Classification of Dynamic Mixers:** We formally define a novel mathematical class of state-conditioned aggregators that move beyond fixed monotonic topologies.
- **Consistency Guarantees:** We prove DIVA satisfies *Dynamic IGM* property under selected assumptions, ensuring decentralized optimality even as the mixing logic reconfigures.
- **Lipschitz Robustness Bounds:** We derive the first suboptimality bound for dynamic value factorization, establishing bound $O\left(\|s - s_0\|\right)$ on the regret caused by state perturbations.

2. Related Works

Under the Centralized Training with Decentralized Execution (CTDE) paradigm (Amato, 2024), coordination in cooperative multi-agent reinforcement learning is commonly achieved through value factorization methods popularized by VDN and QMIX ((Sunehag et al., 2018), (Rashid et al., 2018)). Subsequent methods such as QPLEX (Wang et al., 2021) and QTRAN (Son et al., 2019) further expanded representational capacity through duplex dueling and transformation-based decompositions.More recent MARL approaches incorporate relational and temporal structure to capture inter-agent dependencies. For example, QGNN (Kortvelesy and Prorok, 2022) uses graph neural networks to produce permutation-invariant mixers conditioned on the global state, while TIGER-MARL (Gupta et al., 2025) models the temporal evolution of coordination through dynamic interaction graphs. However, these approaches still rely on a stationary mixing topology: even if conditioned on state or history, they use parameters within a fixed aggregation operator rather than altering the operator itself.

In contrast, many recent MARL architectures introduce state-dependent coordination, allowing agent influence to vary with context through learned interaction structures such as graphs or relational modules (Li et al., 2021; Böhmer et al., 2020; Duan et al., 2024a,b). This ability to adapt coordination structure may explain why such architectures often outperform classical value factorization methods. This motivates introducing dynamism explicitly into mixing functions and therefore making them more competitive within modern MARL frameworks. And consequently shows the need for formalizing theoretical guarantees to preserve decentralized optimality and stability (Zhou et al., 2023; Mei et al., 2023) within dynamic mixing functions.

Existing theoretical analyses of value factorization focus on the Individual–Global Max (IGM) principle under static mixing assumptions (Hong et al., 2022; Chen et al., 2025; Xu et al., 2023), and do not address decentralized optimality or value ordering stability under state-dependent mixing. Moreover, most existing work on robustness in multi-agent reinforcement learning focuses

on policy-level considerations, such as regularization or adversarial training (Shalev-Shwartz et al., 2016; Xu et al., 2023; Li et al., 2019), rather than on the regularity of the value aggregation mechanism. In contrast, DIVA studies robustness directly at the level of the mixing operator. By deriving explicit Lipschitz continuity and regret bounds, we provide guarantees that the transition from static to dynamic mixers remains numerically stable, establishing a linear regret bound $O(\|s - s_0\|)$.

3. Preliminaries

3.1. Cooperative Partially Observable Stochastic Game.

We consider a cooperative partially observable stochastic game (POSG) $\mathcal{M} = \langle \mathcal{D}, \mathcal{S}, \{\mathcal{A}_i\}_{i\in\mathcal{D}}, \mathcal{P}, \mathcal{R}, \{\mathcal{O}_i\}_{i\in\mathcal{D}}, \gamma \rangle$, where $\mathcal{D} = \{1, \dots, N\}$ denotes the set of agents, $\mathcal{S}$ the global state space, $\mathcal{A}_i$ the action space of agent i, and $\mathcal{A} := \prod_{i=1}^{N} \mathcal{A}_i$ the joint action space. At time t, the system transitions according to $\mathcal{P}(s_{t+1} \mid s_t, \mathbf{a}_t)$ and emits a shared reward $\mathcal{R}(s_t, \mathbf{a}_t) \in \mathbb{R}$. Each agent i receives a local observation $o_{t,i} \in \mathcal{O}_i$. Each agent maintains a local history $\tau_{t,i} \in (\mathcal{O}_i \times \mathcal{A}_i)^t$ and follows a decentralized policy $\pi_i(a_{t,i} \mid \tau_{t,i})$. The joint policy is denoted by $\pi = (\pi_1, \dots, \pi_N)$.

The centralized joint action–value function induced by π is

$$Q^{\pi}(s_t, \mathbf{a}_t) := \mathbb{E}_{\pi}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \,\middle|\, s_t, \mathbf{a}_t\right].$$

The optimal joint action–value function is given by

$$Q^*(s, \mathbf{a}) := \max_{\pi} Q^{\pi}(s, \mathbf{a}).$$

3.2. Local Utilities and Value Factorization.

We consider value-decomposition methods that approximate the optimal joint action–value function via a set of per-agent utilities

$$q_i : \mathcal{S} \times \mathcal{A}_i \to \mathbb{R}, \qquad i \in \mathcal{D}.$$

A mixing (aggregation) function $M : \mathcal{S} \times \mathbb{R}^N \to \mathbb{R}$ induces a factorized joint value function

$$Q_{\text{tot}}(s, \mathbf{a}) := M\big(s, q_1(s, a_1), \dots, q_N(s, a_N)\big),$$

which serves as a structured approximation to $Q^*(s, \mathbf{a})$. We use Q_{tot} for centralized training and decentralized greedy action selection.

3.3. Individual–Global Max (IGM).

A value factorization $(\{q_i\}_{i\in\mathcal{D}}, M)$ is said to satisfy the *Individual–Global Max (IGM)* condition if, for all states $s \in \mathcal{S}$, the decentralized greedy joint action coincides with a global maximizer of Q_{tot}:

$$\arg\max_{\mathbf{a}\in\mathcal{A}} Q_{\text{tot}}(s, \mathbf{a}) = \prod_{i=1}^{N} \arg\max_{a_i \in \mathcal{A}_i} q_i(s, a_i).$$

Equivalently, for all $s \in \mathcal{S}$ and all $\mathbf{a} \in \mathcal{A}$,

$$Q_{\text{tot}}(s, \mathbf{u}^*(s)) \geq Q_{\text{tot}}(s, \mathbf{a}),$$

where $\mathbf{u}^*(s)$ denotes the decentralized greedy joint action.

4. Dynamic Individual-Global Max Value Aggregation

4.1. Dynamic Value factorization Class

The fundamental limitation of existing value factorization methods lies in their use of *stationary mixing topologies*. In architectures like QMIX or VDN, the mathematical relationship between agent utilities q_i and the global value Q_{tot} is fixed throughout the state space. While these models may use state information to tune the numerical weights of the mixer, the underlying aggregation logic remains constant.

The Dynamic Mixing Operator We introduce a novel class of Dynamic Mixers where the aggregation operator itself is explicitly reconfigured by the state s. Our framework, Dynamic Individual-Global-Max Value Aggregation (DIVA), employs the state to dynamically adapt the topology of the value factorization.

The global joint action-value function $Q_{\text{tot}}(s, \mathbf{u})$ is defined as a dynamic linear mixture of individual agent utilities q_i:

$$Q_{\text{tot}}(s, \mathbf{u}) := M\Big(s,\ g_1(s)q_1(s, u_1),\ \ldots,\ g_n(s)q_n(s, u_n)\Big) \tag{1}$$

where $q_i : \mathcal{S} \times \mathcal{U}_i \to \mathbb{R}$ is the local utility function and $g_i : \mathcal{S} \to \mathbb{R}$ is the state-dependent gating function of global state s.

To facilitate the analysis of local stability, we define the gated utility for each agent i as the state-modulated contribution to the global value:

$$x_i(s, u_i) := g_i(s) \cdot q_i(s, u_i). \tag{2}$$

This structure allows agent importance and interactions to change as the environment evolves, since the aggregation operator is explicitly conditioned on the global state. As a result, the coordination logic depends directly on the state s, rather than emerging implicitly from fixed mixing weights or architectural constraints.

4.2. Structural Requirements

Before establishing formal guarantees for dynamic value factorization, we identify a set of structural conditions under which state-dependent mixing functions are well-defined and amenable to analysis. These conditions restrict the regularity and ordering properties of the mixing architecture, ensuring that variations in the global state induce controlled changes in the aggregated value function. The assumptions introduced in this section are sufficient to guarantee decentralized optimality and robustness results as derived in the following sections. They serve as a foundational blueprint for translating the abstract principles of DIVA into implementable mixing architectures, as shown in Section 6.

Boundedness Dynamic aggregation introduces the risk that unbounded utilities or mixing weights may amplify small perturbations in the state into arbitrarily large changes in the joint value function. To ensure that the induced value landscape remains well-posed locally, we impose a boundedness condition on both agent utilities and gating functions.

Assumption 1 (Local Boundedness) *There exist constants $B_q, B_g > 0$ and a radius $r_b > 0$ such that for all $s \in B_{r_b}(s_0)$ we have $|q_i(s, u_i)| \leq B_q$ and $|g_i(s)| \leq B_g$.*

Smoothness (Lipschitz Continuity) Moreover, state-dependent aggregation induces a family of joint value functions Q_{tot} indexed by the global state. To reason about decentralized execution, it is necessary to understand how the ordering of joint actions under Q_{tot} changes as s varies. Without quantitative control, an arbitrarily small perturbation of the state may induce a large change in the value assigned to a fixed joint action, potentially reordering action values and breaking local optimality arguments.

Lipschitz continuity (Appendix A.2) provides a minimal regularity condition that bounds these variations linearly in the state perturbation. Such bounds allow us to relate changes in the state directly to changes in joint action values, which is sufficient to establish stability of greedy action selection and derive robustness guarantees under dynamic mixing.

Assumption 2 (Smoothness/Local Lipschitz Continuity) *There exist constants $L_q, L_g, L_M, L_{M,s} > 0$ and a radius $r_L > 0$ such that for all $s, s' \in B_{r_L}(s_0)$:*

1. ***(Utilities)*** $|q_i(s, u_i) - q_i(s', u_i)| \leq L_q \|s - s'\|$
2. ***(Gates)*** $|g_i(s) - g_i(s')| \leq L_g \|s - s'\|$
3. ***(Mixer inputs)*** $|M(s, \mathbf{x}) - M(s, \mathbf{y})| \leq L_M \|x - y\|_2$
4. ***(Mixer state)*** $|M(s, \mathbf{x}) - M(s', \mathbf{x})| \leq L_{M,s} \|s - s'\|$

Consistency and Monotonicity Finally, in a dynamic setting, if a gating weight were allowed to become non-positive, an agent's local maximization could be ignored or even inverted by the global mixer, violating the IGM principle. To guarantee that the decentralized greedy action $\mathbf{u}^*(s)$ always corresponds to a global optimum, we utilize gating functions that ensure strict positivity and maintain a monotonic relationship between agent utilities and the joint value estimate.

Assumption 3 (Gate Positivity) *There exists a radius $r_+ > 0$ such that for all agents i and all states $s \in B_{r_+}(s_0)$, the gating function is strictly positive: $g_i(s) > 0$.*

Strict positivity of the gating functions ensures that the relative ordering of each agent's local utilities is preserved under gating, so that maximizing $q_i(s, \cdot)$ is equivalent to maximizing the gated contribution $g_i(s)q_i(s, \cdot)$. However, positivity alone does not constrain how the aggregation function combines these gated utilities. Without additional structure, the mixer could introduce interactions that reverse component-wise improvements, allowing increases in individual utilities to decrease the joint value estimate. To prevent such order inversions, we impose a monotonicity condition on the mixing function.

Assumption 4 (Monotonicity in Gated Utilities) *For each fixed $s \in \mathcal{S}$, the map $x \mapsto M(s, x)$ is coordinate-wise non-decreasing: if $x \leq y$ component-wise, then $M(s, x) \leq M(s, y)$.*

5. Theoretical Analysis and Performance Guarantees

This section develops a sequence of technical lemmas that characterize how state-dependent gating affects the regularity and ordering properties of the induced joint value function. The results establish (i) Lipschitz continuity of the gated utilities and the aggregated value function with respect to

the global state, and (ii) equivalence between local and gated optimization problems under positivity. Together, these properties will aid in proving dynamic IGM guarantees and robustness results in the following section.

We begin by analyzing the effect of state-dependent gating at the level of individual agents. Since the gated utility $x_i(s, u_i) = g_i(s)q_i(s, u_i)$ combines two state-dependent components, its regularity is not immediate from the regularity of q_i and g_i alone. The following lemma shows that, under boundedness and Lipschitz assumptions, each gated coordinate remains Lipschitz continuous in the state.

Lemma 1 (Lipschitz continuity of each gated coordinate) *Assume 1 and the utility/gate Lipschitz parts of 2. Let*

$$r_x := \min\{r_b, r_L\}.$$

Then for every i and every $u_i \in \mathcal{U}_i$, the function $s \mapsto x_i(s, u_i)$ is Lipschitz on $B_{r_x}(s_0)$ with constant

$$L_x := B_g L_q + B_q L_g,$$

i.e. for all $s, s' \in B_{r_x}(s_0)$,

$$|x_i(s, u_i) - x_i(s', u_i)| \leq L_x \left\|s - s'\right\|.$$

Proof Fix i and u_i and abbreviate $q_i(s) := q_i(s, u_i)$. Then

$$x_i(s, u_i) - x_i(s', u_i) = g_i(s)q_i(s) - g_i(s')q_i(s').$$

We can add and subtract $g_i(s)q_i(s')$ to RHS to obtain:

$$g_i(s)\big(q_i(s) - q_i(s')\big) + q_i(s')\big(g_i(s) - g_i(s')\big)$$

Taking absolute values and using the triangle inequality (Appendix A.4) we get:

$$|x_i(s, u_i) - x_i(s', u_i)| \leq$$

$$|g_i(s)|\,|q_i(s) - q_i(s')| + |q_i(s')|\,|g_i(s) - g_i(s')|.$$

Note $s, s' \in B_{r_x}(s_0) \subseteq B_{r_b}(s_0)$, thus Assumption 1 gives $|g_i(s)| \leq B_g$ and $|q_i(s')| \leq B_q$. Similarly $s, s' \in B_{r_x}(s_0) \subseteq B_{r_L}(s_0)$, and hence Assumption 2 gives $|q_i(s) - q_i(s')| \leq L_q\|s - s'\|$ and $|g_i(s) - g_i(s')| \leq L_g\|s - s'\|$. Therefore,

$$|x_i(s, u_i) - x_i(s', u_i)| \leq (B_g L_q + B_q L_g)\left\|s - s'\right\|$$

■

Having established Lipschitz continuity of each gated coordinate, we now lift this property to the aggregated joint value function by combining the coordinate-wise bounds with the regularity of the mixing function.

Lemma 2 (Lipschitz continuity of $Q_{\text{tot}}(\cdot, \mathbf{u})$) *Assume 1 and 2. Let*

$$r := \min\{r_b, r_L\}.$$

Then for every fixed $\mathbf{u} \in \mathcal{U}$ and all $s, s' \in B_r(s_0)$,

$$|Q_{\text{tot}}(s, \mathbf{u}) - Q_{\text{tot}}(s', \mathbf{u})| \leq K_{\text{tot}} \left\|s - s'\right\|,$$

where

$$K_{\text{tot}} := L_M \sqrt{n}\, L_x + L_{M,s}, \qquad L_x = B_g L_q + B_q L_g.$$

Proof Fix $\mathbf{u} = (u_1, \ldots, u_n)$. Write

$$Q_{\text{tot}}(s, \mathbf{u}) - Q_{\text{tot}}(s', \mathbf{u}) = M\big(s, x(s, \mathbf{u})\big) - M\big(s', x(s', \mathbf{u})\big).$$

We add and subtract $M\big(s, x(s', \mathbf{u})\big)$ to RHS and once more use the Triangle Inequality (Appendix A.4) to get

$$|Q_{\text{tot}}(s, \mathbf{u}) - Q_{\text{tot}}(s', \mathbf{u})| \leq \Big|M\big(s, x(s, \mathbf{u})\big) - M\big(s, x(s', \mathbf{u})\big)\Big| + \Big|M\big(s, x(s', \mathbf{u})\big) - M\big(s', x(s', \mathbf{u})\big)\Big|.$$

For $s \in B_r(s_0) \subseteq B_{r_L}(s_0)$, the mixer-input Lipschitz property gives

$$\Big|M\big(s, x(s, \mathbf{u})\big) - M\big(s, x(s', \mathbf{u})\big)\Big| \leq L_M \left\|x(s, \mathbf{u}) - x(s', \mathbf{u})\right\|_2.$$

Also, for $s, s' \in B_r(s_0) \subseteq B_{r_L}(s_0)$, the mixer-state Lipschitz property gives

$$\Big|M\big(s, x(s', \mathbf{u})\big) - M\big(s', x(s', \mathbf{u})\big)\Big| \leq L_{M,s} \left\|s - s'\right\|.$$

Hence it remains to bound $\|x(s, \mathbf{u}) - x(s', \mathbf{u})\|_2$.
By Lemma 1, for each coordinate i,

$$|x_i(s, u_i) - x_i(s', u_i)| \leq L_x \left\|s - s'\right\|.$$

Therefore,

$$\left\|x(s, \mathbf{u}) - x(s', \mathbf{u})\right\|_2^2 = \sum_{i=1}^{n} |x_i(s, u_i) - x_i(s', u_i)|^2 \leq \sum_{i=1}^{n} (L_x \left\|s - s'\right\|)^2 = n L_x^2 \left\|s - s'\right\|^2$$

so $\|x(s, \mathbf{u}) - x(s', \mathbf{u})\|_2 \leq \sqrt{n}\, L_x \|s - s'\|$. Combining the bounds yields

$$|Q_{\text{tot}}(s, \mathbf{u}) - Q_{\text{tot}}(s', \mathbf{u})| \leq \big(L_M \sqrt{n} L_x + L_{M,s}\big) \left\|s - s'\right\|$$

■

While the above Lemmas aid in controlling how the joint value function varies with the state, the next result describes how state-dependent gating affects the local optimization problems solved by individual agents. We will show that, under gate positivity, maximizing the raw utility is equivalent to maximizing the gated utility, ensuring that local greedy actions are well-defined under dynamic aggregation.

Lemma 3 (Equivalence of Local Objectives) *Assume Assumption 3 (Gate Positivity). Let r_+ be the radius of positivity. For all states $s \in B_{r_+}(s_0)$, the set of maximizers for the raw utility q_i is identical to the set of maximizers for the gated utility $x_i(s, u_i) = g_i(s)q_i(s, u_i)$.*

Formally, for any agent i and any action $u_i^* \in \mathcal{U}_i$:

$$u_i^* \in \arg\max_{u_i \in \mathcal{U}_i} q_i(s, u_i) \iff u_i^* \in \arg\max_{u_i \in \mathcal{U}_i} \Big(g_i(s)q_i(s, u_i) \Big).$$

Proof Fix any state $s \in B_{r_+}(s_0)$. By Assumption 3, we have $g_i(s) > 0$.
($\Rightarrow$) Suppose u_i^* maximizes $q_i(s, \cdot)$. Then for all $u_i \in \mathcal{U}_i$:

$$q_i(s, u_i^*) \geq q_i(s, u_i).$$

Since $g_i(s)$ is a strictly positive scalar, we can multiply both sides of the inequality by $g_i(s)$ without changing the direction of the inequality:

$$g_i(s)q_i(s, u_i^*) \geq g_i(s)q_i(s, u_i).$$

By the definition of gated utility, we get $x_i(s, u_i^*) \geq x_i(s, u_i)$. Thus, u_i^* is a maximizer of $x_i(s, \cdot)$.
($\Leftarrow$) Conversely, suppose u_i^* maximizes $x_i(s, \cdot)$. Then:

$$g_i(s)q_i(s, u_i^*) \geq g_i(s)q_i(s, u_i).$$

Dividing both sides by $g_i(s)$ (which is valid since $g_i(s) \neq 0$) recovers the original inequality:

$$q_i(s, u_i^*) \geq q_i(s, u_i).$$

Thus, the sets of maximizers are identical. ■

These results allow us to treat gated utilities as order-preserving transformations of local utilities and to control the variation of Q_{tot} with respect to the state, which are the key ingredients for establishing dynamic IGM consistency. We can now move on to establishing two the central theoretical contributions of this work.

Theorem: Dynamic IGM Consistency

Calssical IGM condition guarantees decentralized optimality only for value factorizations with state-independent mixing functions. In contrast, state-dependent mixing introduces a family of aggregation operators whose structure varies with the environment, for which classical IGM arguments no longer apply. We show that, under strict positivity and monotonicity, decentralized greedy action selection remains globally optimal despite this dynamic reconfiguration, thereby extending IGM consistency to a broad class of state-conditioned value factorizations.

To rigorously prove consistency, we define the solution sets for the global and local problems.

Definition 4 (Optimal and Greedy Action Sets) *For any state $s \in \mathcal{S}$:*

1. *Let $\mathcal{U}^*_{glob}(s)$ denote the set of **global joint maximizers**:*

$$\mathcal{U}^*_{glob}(s) := \left\{ \mathbf{u} \in \mathcal{U} \;\middle|\; Q_{\text{tot}}(s, \mathbf{u}) = \max_{\mathbf{u}' \in \mathcal{U}} Q_{\text{tot}}(s, \mathbf{u}') \right\}.$$

2. *Let $\mathcal{U}^*_{loc}(s)$ denote the set of* ***decentralized greedy joint actions****, defined as the Cartesian product of individual best responses:*

$$\mathcal{U}^*_{loc}(s) := \prod_{i=1}^{n} \left(\arg\max_{u_i \in \mathcal{U}_i} q_i(s, u_i) \right).$$

Theorem 5 (Decentralized Optimality / IGM Consistency) *Assume Assumption 3 (Positivity) and Assumption 4 (Monotonicity). For all states $s \in B_{r_+}(s_0)$, the set of decentralized greedy strategies is a subset of the global optimal strategies:*

$$\mathcal{U}^*_{loc}(s) \subseteq \mathcal{U}^*_{glob}(s).$$

Consequently, any joint action $\mathbf{u}^*$ constructed by independent greedy agents is guaranteed to maximize the global objective Q_{tot}.

Proof We proceed by direct construction of the inequality chain. Fix any state $s \in B_{r_+}(s_0)$.
Step 1: Selection of Arbitrary Candidates. Let $\mathbf{u}^* = (u_1^*, \ldots, u_n^*) \in \mathcal{U}^*_{loc}(s)$ be any arbitrary decentralized greedy joint action. Let $\mathbf{u} = (u_1, \ldots, u_n) \in \mathcal{U}$ be any arbitrary comparison joint action.
Step 2: Component-wise Dominance. We invoke Lemma 3 (Equivalence of Local Objectives). Since each component u_i^* is a maximizer of the raw utility $q_i(s, \cdot)$, Lemma 3 ensures it is also a maximizer of the gated input $x_i(s, \cdot)$. Therefore, for every agent $i \in \{1, \ldots, n\}$ and any alternative action u_i we have $x_i(s, u_i^*) \geq x_i(s, u_i)$. Aggregating these inequalities over all n agents implies:

$$x(s, \mathbf{u}^*) \geq x(s, \mathbf{u}). \tag{3}$$

Step 3: Monotonicity Preservation. We invoke Assumption 4. The mixing function $M(s, \cdot)$ is monotonically non-decreasing in its arguments. Applying the mixer $M(s, \cdot)$ to both sides of inequality (3) preserves the order:

$$M\big(s, x(s, \mathbf{u}^*)\big) \geq M\big(s, x(s, \mathbf{u})\big).$$

By the definition of the total action-value function Q_{tot} (Eq. 1), this is exactly $Q_{\text{tot}}(s, \mathbf{u}^*) \geq Q_{\text{tot}}(s, \mathbf{u})$.
Step 4: Conclusion of Global Optimality. We have shown that $Q_{\text{tot}}(s, \mathbf{u}^*) \geq Q_{\text{tot}}(s, \mathbf{u})$ for an arbitrary $\mathbf{u} \in \mathcal{U}$. It follows by definition that $\mathbf{u}^*$ achieves the maximum possible value, thus $Q_{\text{tot}}(s, \mathbf{u}^*) = \max_{\mathbf{u}' \in \mathcal{U}} Q_{\text{tot}}(s, \mathbf{u}')$ Hence, $\mathbf{u}^* \in \mathcal{U}^*_{glob}(s)$. Since $\mathbf{u}^*$ was an arbitrary element of $\mathcal{U}^*_{loc}(s)$, the inclusion $\mathcal{U}^*_{loc}(s) \subseteq \mathcal{U}^*_{glob}(s)$ holds. ■

Theorem: Value Continuity and Robustness

While IGM consistency guarantees alignment between decentralized greedy decisions and the global objective at a fixed state, real-world multi-agent systems operate under state uncertainty, sensing noise, and delayed updates. In such settings, value factorization methods must be stable to small state perturbations. The following result formalizes this requirement by establishing Lipschitz continuity of the optimal value function and by bounding the regret incurred when executing a reference-optimal joint action under state changes. These bounds imply that dynamic value factorization does not require continuous re-optimization to remain effective. The following theorem provides a principled robustness analysis and derives Lipschitz guarantees for decentralized execution.

Theorem 6 (Value Lipschitz continuity and old-action regret bound) *Assume 1 and 2. Let*

$$R := \min\{r_b, r_L\},$$

and let K_{tot} be as in Lemma 2. Define the optimal value function

$$V^*(s) := \max_{\mathbf{u}\in\mathcal{U}} Q_{\text{tot}}(s, \mathbf{u}).$$

Then for all $s \in B_R(s_0)$:

(A) ***(Lipschitz continuity of the optimal value)***

$$\left|V^*(s) - V^*(s_0)\right| \le K_{\text{tot}} \|s - s_0\|.$$

(B) ***(Robustness of retaining the reference-optimal action)*** *Let $\mathbf{u}^\dagger \in \arg\max_{\mathbf{u}\in\mathcal{U}} Q_{\text{tot}}(s_0, \mathbf{u})$ (not necessarily unique). Then the regret of executing $\mathbf{u}^\dagger$ at the perturbed state s is bounded:*

$$0 \le V^*(s) - Q_{\text{tot}}(s, \mathbf{u}^\dagger) \le 2K_{\text{tot}} \|s - s_0\|.$$

Proof Fix any $s \in B_R(s_0)$.
Step 1 (Lipschitz bounds for fixed actions). By Lemma 2, for every fixed joint action $\mathbf{u} \in \mathcal{U}$,

$$\left|Q_{\text{tot}}(s, \mathbf{u}) - Q_{\text{tot}}(s_0, \mathbf{u})\right| \le K_{\text{tot}} \|s - s_0\|.$$

Equivalently for all $\mathbf{u} \in \mathcal{U}$:,

$$Q_{\text{tot}}(s_0, \mathbf{u}) - K_{\text{tot}} \|s - s_0\| \;\le\; Q_{\text{tot}}(s, \mathbf{u}) \;\le\; Q_{\text{tot}}(s_0, \mathbf{u}) + K_{\text{tot}} \|s - s_0\| \tag{4}$$

Step 2 (prove (A):bound $V^*(s)$). Taking the maximum over $\mathbf{u} \in \mathcal{U}$ in RHS of (4) yields $V^*(s) \le V^*(s_0) + K_{\text{tot}} \|s - s_0\|$. Similarly, taking the maximum over $\mathbf{u} \in \mathcal{U}$ in LHS of (4) gives $V^*(s) \ge V^*(s_0) - K_{\text{tot}} \|s - s_0\|$. Combining both inequalities finishes proof of part (A)

$$\left|V^*(s) - V^*(s_0)\right| \le K_{\text{tot}} \|s - s_0\|,$$

Step 3 (prove (B): nonnegativity of the regret). By definition of $V^*(s)$ as a maximum,

$$V^*(s) = \max_{\mathbf{u}} Q_{\text{tot}}(s, \mathbf{u}) \;\ge\; Q_{\text{tot}}(s, \mathbf{u}^\dagger),$$

so $V^*(s) - Q_{\text{tot}}(s, \mathbf{u}^\dagger) \ge 0$.
Step 4 (prove (B): upper bound on the regret). From Step 2: $V^*(s) \le V^*(s_0) + K_{\text{tot}} \|s - s_0\|$. Also, applying LHS of (4) to the specific action $\mathbf{u}^\dagger$ gives $Q_{\text{tot}}(s, \mathbf{u}^\dagger) \ge Q_{\text{tot}}(s_0, \mathbf{u}^\dagger) - K_{\text{tot}} \|s - s_0\|$. Subtracting the latter from the former yields

$$V^*(s) - Q_{\text{tot}}(s, \mathbf{u}^\dagger) \le \Big(V^*(s_0) + K_{\text{tot}} \|s - s_0\|\Big) - \Big(Q_{\text{tot}}(s_0, \mathbf{u}^\dagger) - K_{\text{tot}} \|s - s_0\|\Big)$$

Finally, since $\mathbf{u}^\dagger \in \arg\max_{\mathbf{u}} Q_{\text{tot}}(s_0, \mathbf{u})$, we have $V^*(s_0) = Q_{\text{tot}}(s_0, \mathbf{u}^\dagger)$, and therefore

$$V^*(s) - Q_{\text{tot}}(s, \mathbf{u}^\dagger) \le 2K_{\text{tot}} \|s - s_0\|.$$

■

Together, the two results characterize complementary aspects of dynamic value factorization. Theorem 5 establishes *consistency*, showing that decentralized greedy execution remains globally optimal under state-dependent mixing. Theorem 6 establishes *robustness*, quantifying how deviations from instantaneous policy updates translate into bounded performance loss under state perturbations. This validates DIVA as a reliable value factorization framework, preserving decentralized optimality under dynamic mixing.

6. Certifying Dynamic Value Factorization

Having established abstract consistency and robustness properties for dynamic value factorization, we now examine whether these guarantees can be realized by concrete state-dependent mixing architectures. We introduce representative classes of dynamic mixers and verify that standard bounded agent utilities and hypernetwork-based gating mechanisms satisfy the structural assumptions required by our analysis.

6.1. State-Dependent Mixing Architectures

We consider architectures where the joint action–value function is formed by dynamically aggregating per-agent utilities according to the global state:

$$Q_{\text{tot}}(s, \mathbf{u}) = \sum_{i=1}^{n} g_i(s)\, q_i(s, u_i),$$

where the gating functions $g_i(s)$ are generated by a hypernetwork W_h conditioned on s. To ensure Lipschitz stability, we enforce a spectral normalization constraint $\|W_h\|_2 \leq 1$ on the hypernetwork weights.

Softmax variant We initialize gating functions via a Softmax transformation on the logits $z = W_h s$:

$$g_i(s) = \frac{e^{[W_h s]_i}}{\sum_{j=1}^{n} e^{[W_h s]_j}}.$$

This construction yields a normalized, state-dependent aggregation in which relative agent contributions are modulated by the environment.

Softplus variant Gating functions are generated using a Softplus activation with an offset $\epsilon > 0$:

$$g_i(s) = \ln\Big(1 + e^{[W_h s]_i}\Big) + \epsilon.$$

Unlike Softmax gating, this formulation does not impose a normalization constraint and allows multiple agents to simultaneously exert strong influence on the aggregated value.

In both variants, each agent utility q_i utilizes an RNN with spectral normalization ($\|W\|_2 \leq 1$) and a scaled hyperbolic tangent activation:

$$q_i(s, u_i) = R_{\max} \cdot \tanh(\phi_i(s, u_i; \theta_i)).$$

6.2. Verification of Structural Assumptions

We verify that these architectures satisfy the theoretical requirements for Dynamic IGM consistency.

Local boundedness. Assumption 1 requires that utilities and gates remain bounded in a neighborhood of s_0. Boundedness of agent utilities follows from the scaled activation. For the Softmax variant, we have $\sum_i g_i(s) = 1$ and thus gating functions g_i are globally bounded by $B_g = 1$. For Softplus, since the mapping $s \mapsto W_h s$ is continuous and the neighborhood $B_{r_b}(s_0)$ is compact, the Extreme Value Theorem (Appendix A.5) guarantees a finite maximum $B_g < \infty$:

$$B_g := \max_i \sup_{s \in B_{r_b}(s_0)} |\ln(1 + \exp([W_h s]_i)) + \epsilon| < \infty.$$

Lipschitz continuity. Assumption 2 requires that small perturbations in the state or agent utilities do not lead to disproportionate changes in the joint value function. For differentiable functions, the Lipschitz constant is defined as the supremum of the gradient norm: $L = \sup_s \|\nabla f(s)\|$ (Appendix A.8). We verify this property by bounding the derivatives of each component.

1. **Agent Stability (L_q):** The agent utility is defined as $q_i = R_{\max} \tanh(\phi_i)$. Let $f(x) = \tanh(x)$; its derivative $f'(x) = 1 - \tanh^2(x)$ is bounded by $|f'(x)| \leq 1$. By the chain rule (Appendix A.6), the gradient with respect to the state is $\nabla_s q_i = R_{\max} \cdot f'(\cdot) \cdot W$. Since the weights are constrained by spectral normalization ($\|W\|_2 \leq 1$), the total Lipschitz constant is bounded by the scaling factor:$L_q \leq R_{\max} \cdot \sup |f'| \cdot \|W\|_2 \leq R_{\max}$

2. **Gate Stability (L_g):** The gating weights $g_i(s)$ are compositions of a hypernetwork W_h and an activation function. The Softmax function is known to be 1-Lipschitz with respect to the ℓ_2 norm. For the Softplus variant, the derivative is the sigmoid function $\sigma(z) = \frac{1}{1+e^{-z}}$, bounded by $(0, 1)$. In both cases, $\|W_h\|_2 \leq 1$ ensures that the gating weights vary smoothly:
$L_g \leq \sup |\sigma| \cdot \|W_h\|_2 \leq 1$

3. **Mixing Input Stability (L_M):** The stability of the global value w.r.t. agent utility inputs is determined by the weighting vector $\mathbf{g}(s)$. Since the mixer is linear in $\mathbf{q}$, the gradient is $\nabla_{\mathbf{q}} M = \mathbf{g}(s)$. The Lipschitz constant is thus $L_M = \sup_s \|\mathbf{g}(s)\|_2$. For the Softmax variant we have $\|\mathbf{g}\|_1 = 1$, which implies $\|\mathbf{g}\|_2 \leq 1$. Thus, $L_M \leq 1$. For the Softplus variant, since $\mathbf{g}(s)$ is locally bounded by B_g, it follows that $L_M \leq B_g$.

4. **Structural Sensitivity ($L_{M,s}$):** We analyze the sensitivity of the mixing structure to state changes while holding utilities constant. Applying the product rule (Appendix A.6), the structural gradient is $\nabla_s^{\text{struct}} M = (\nabla_s \mathbf{g})^T \mathbf{q}$. By the Cauchy-Schwarz inequality (Appendix A.7) and our established bounds: $L_{M,s} \leq \|\nabla_s \mathbf{g}\| \cdot \|\mathbf{q}\| \leq L_g \cdot B_q = R_{\max}$

Strict positivity and Monotonicity. Assumptions 3 and 4 hold by construction. Since $e^x > 0$ and $\ln(1 + e^x) + \epsilon \geq \epsilon > 0$, all weights $g_i(s)$ are strictly positive. Furthermore, since $\frac{\partial Q_{\text{tot}}}{\partial q_i} = g_i(s) > 0$, the joint value increases monotonically with agent utilities, satisfying the IGM condition.

7. Conclusion

In this paper, we introduced DIVA, a rigorous theoretical framework validating a new class of state-dependent mixing functions in cooperative MARL. By identifying a set of four sufficient structural conditions, we have provided the first formal methodology for moving beyond stationary mixing topologies while maintaining the foundational Individual–Global Max (IGM) principle.Moreover, our contribution also lies in the derivation of the first formal theoretical guarantees for suboptimality in dynamic value factorization. Specifically, our established linear regret bound $O(\|s - s_0\|)$ ensures that architectural flexibility does not compromise stability. Furthermore, our verification of common gating mechanisms like Softmax and Softplus certifies that this framework is immediately applicable.

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Appendix A. Extended Mathematical Preliminaries

This section provides a rigorous foundation for the analytical results presented in the main text. We expand upon the definitions of regularity, monotonicity, and the algebraic inequalities required to certify the robustness of the **DIVA** framework.

A.1. Norms, Neighborhoods, and Local Regularity

We assume the state space $\mathcal{S} \subseteq \mathbb{R}^d$ is a normed vector space equipped with a norm $\|\cdot\|$. In high-dimensional reinforcement learning tasks, global functions may not be uniformly smooth; thus, we focus on local regularity.

Definition 7 (Open Ball) *For a reference state $s_0 \in \mathcal{S}$ and a radius $r > 0$, the open ball is defined as:*

$$B_r(s_0) := \{s \in \mathcal{S} : \|s - s_0\| < r\}.$$

A.2. Lipschitz Continuity and Sensitivity

Lipschitz continuity provides a formal measure of a system's sensitivity to input perturbations or sensor noise.

Definition 8 (Lipschitz Continuity) *A function $f : \mathcal{X} \to \mathcal{Y}$ between normed spaces is L-Lipschitz on a domain $D \subseteq \mathcal{X}$ if:*

$$\|f(x) - f(x')\|_{\mathcal{Y}} \leq L\|x - x'\|_{\mathcal{X}} \quad \forall x, x' \in D.$$

The constant L represents the maximum gain of the system. In the context of

A.3. Order Preservation and Monotonicity

To ensure that decentralized optimization remains consistent with global objectives, the mixing function must preserve the relative ordering of agent preferences.

Definition 9 (Coordinate-wise Monotonicity) *A function $M : \mathbb{R}^n \to \mathbb{R}$ is coordinate-wise non-decreasing if for all $x, y \in \mathbb{R}^n$:*

$$x \leq y \quad \Rightarrow \quad M(x) \leq M(y),$$

where $x \leq y$ denotes $x_i \leq y_i$ for all $i \in \{1, \ldots, n\}$.

A.4. Auxiliary Inequalities

We collect the specific variants of the triangle inequality and product perturbation bounds used to derive the Lipschitz constants in Lemma 1 and Theorem 6.

Lemma 10 (Standard and Reverse Triangle Inequality) *Let V be a normed vector space. For any $x, y \in V$:*

1. ***Standard:*** $\|x + y\| \leq \|x\| + \|y\|$.
2. ***Reverse:*** $\big|\|x\| - \|y\|\big| \leq \|x - y\|$.

Proof The standard form follows directly from the definition of a norm. For the reverse form, note that:

$$\|x\| = \|x - y + y\| \leq \|x - y\| + \|y\|$$

which implies $\|x\| - \|y\| \leq \|x - y\|$. Swapping x and y and using $\|x - y\| = \|y - x\|$ completes the proof. ■

Lemma 11 (Product Perturbation Bound) *Let $a, a', b, b' \in \mathbb{R}$. If $|a| \leq B_a$ and $|b'| \leq B_{b'}$, then:*

$$|ab - a'b'| \leq B_a|b - b'| + B_{b'}|a - a'|.$$

Proof Add and subtract the cross-term ab' and apply the triangle inequality:

$$\begin{aligned} |ab - a'b'| &= |a(b - b') + b'(a - a')| \\ &\leq |a|\,|b - b'| + |b'|\,|a - a'|. \end{aligned}$$

Applying the stated bounds yields the result. ■

A.5. Extreme Value Theorem

The Extreme Value Theorem is used to certify boundedness of state-dependent gating functions on local neighborhoods of the state space.

Theorem 12 (Extreme Value Theorem) *Let $K \subseteq \mathbb{R}^d$ be compact and let $f : K \to \mathbb{R}$ be continuous. Then f attains both a maximum and a minimum on K.*

A.6. Differential Calculus of Multi-Agent Mixers

The following rules enable the propagation of derivative and Lipschitz bounds through the compositions and products that define state-dependent value factorization.

Lemma 13 (Chain Rule) *Let $f : \mathbb{R}^m \to \mathbb{R}^k$ and $g : \mathbb{R}^d \to \mathbb{R}^m$ be differentiable. Then $h = f \circ g$ is differentiable and its Jacobian is given by:*

$$J_h(x) = J_f(g(x))\, J_g(x).$$

Lemma 14 (Product Rule) *Let $g(s)$ and $q(s)$ be differentiable scalar functions of the state s. The gradient of their product is given by:*

$$\nabla_s\big(g(s)q(s)\big) = q(s)\nabla_s g(s) + g(s)\nabla_s q(s).$$

A.7. Cauchy–Schwarz Inequality

The Cauchy–Schwarz inequality is a fundamental result used to bound inner products and norm interactions in Euclidean spaces.

Lemma 15 (Cauchy–Schwarz Inequality) *Let $x, y \in \mathbb{R}^d$ be vectors equipped with the Euclidean norm. Then:*

$$|\langle x, y\rangle| \le \|x\|_2\, \|y\|_2.$$

Proof If $y = 0$, the inequality holds trivially. Assume $y \neq 0$ and consider the function

$$f(t) = \|x - ty\|_2^2 \quad \text{for } t \in \mathbb{R}.$$

Since norms are non-negative, $f(t) \ge 0$ for all t. Expanding the square yields

$$f(t) = \|x\|_2^2 - 2t\langle x, y\rangle + t^2\|y\|_2^2.$$

This is a quadratic polynomial in t with non-negative values for all t, which implies that its discriminant is non-positive:

$$(2\langle x, y\rangle)^2 - 4\|x\|_2^2\|y\|_2^2 \le 0.$$

Rearranging gives

$$\langle x, y\rangle^2 \le \|x\|_2^2\, \|y\|_2^2.$$

Taking square roots on both sides completes the proof. ■

A.8. Gradient-Based Characterization of Lipschitz Continuity

For differentiable functions on Euclidean domains, Lipschitz continuity can be certified via uniform bounds on gradient norms.

Lemma 16 (Gradient Bound Implies Lipschitz Continuity) *Let $f : \mathcal{S} \subseteq \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable on a convex domain $\mathcal{S}$. If*

$$\sup_{s\in\mathcal{S}} \|\nabla f(s)\|_2 \le L,$$

then f is L-Lipschitz on $\mathcal{S}$.

Proof The result follows by integrating the gradient along the line segment between any two points and applying the Cauchy–Schwarz inequality. ■